

Google Developer Groups

Management Portal - Database Schema

GDG On Campus . HIT

This document outlines the complete database models and schema architecture for the GDG Management Portal.

1. Usermodel

Field Name	Type	Description & Requirement Reference
_id	String	MongoDB Primary Key (Normalized). Unique identifier for the user. (Required)
name	String	Full name of the member.
email	String	Contact email address.
phone	String	Contact phone number.
password	String (Hashed)	For Email/Password login.
auth_provider	String	Authentication source, e.g., 'local', 'google'.
profile_pic	String (URL)	Profile picture URL.
domain	String	Assigned technical domain, e.g.,

WEB_DEV, APP_DEV.		
role_id	String	Reference to RoleModel for dynamic IAM RBAC
batch	String	Graduation batch year for filtering.
status	String	'active', 'alumni', or 'inactive'.
task_count	Number	Count of lifetime tasks completed.
volatile_data	JSON	Flexible storage for frequently changing metadata.
reset_token	String	Secure token for Forgot Password flow.
reset_token_expiry	DateTime	Expiry timestamp for the reset token.
created_at	DateTime	Timestamp of account creation.
updated_at	DateTime	Timestamp of last profile update.

2. RoleModel (IAM RBAC)

Field Name	Type	Description & Requirement Reference
_id	String	MongoDB Primary Key (Normalized). (e.g., 'ROLE_DOMAIN_HEAD')
name	String	Human-readable role name.
description	String	Brief explanation of the role's purpose.
policies	Array[String]	List of associated Policy IDs.

3. PolicyModel (IAM RBAC)

Field Name	Type	Description & Requirement Reference
_id	String	MongoDB Primary Key. (e.g., 'POLICY_DOMAIN_MANAGEMENT')
version	String	Policy version.
statements	Array[JSON]	Format: { effect, actions[], resource }. Allows dynamic interpolation.

4. Taskmodel

Field Name	Type	Description & Requirement Reference
_id	String	MongoDB Primary Key (Normalized).
name	String	Title of the assigned task.
domain_id	String	Associated technical domain.
description	String	Detailed instructions for the task.
type	String	'team' or 'individual'.
priority	String	e.g., 'high', 'medium', 'low'.
created_by	String	User ID of the task creator.
assigned_to	Array[String]	Array of user IDs assigned to this task.
mentor_ids	Array[String]	Array of User IDs for supervising mentors.

deadline	DateTime	Task submission deadline.
status	String	'not_started', 'in_progress', 'completed', etc.
submissions	Array[JSON]	Format: [{ submitted_by, submission_link, submitted_at }].
reviews	Array[JSON]	Format: [{ reviewed_by, remarks, rating }].
history	Array[JSON]	Audit log of creations, reassignments, etc.
attachments	Array[JSON]	Format: [{ name, url }] for attached files.
created_at	DateTime	Timestamp of creation.

5. Meetingmodel

Field Name	Type	Description & Requirement Reference
_id	String	MongoDB Primary Key (Normalized).
title	String	Title of the meeting.
agenda	String	Meeting agenda and discussion points.
date_time	DateTime	Scheduled date and time.
venue	JSON	Format: { type: 'online' 'offline', location: string null, link: string null }.
target_batch	Array[String]	Target batch years for filtering.

target_domain	Array[String]	Target domains for filtering.
status	String	'upcoming', 'ongoing', 'completed'.
mom_content	String	Minutes of Meeting notes.
created_by	String	Admin/User ID who created the meeting.

6. LeaveApplicationModel

Field Name	Type	Description & Requirement Reference
_id	String	MongoDB Primary Key (Normalized).
user_id	String	User ID of the applicant.
leave_type	String	'period', 'meeting', or 'task'.
target	JSON	{ type: 'meeting' 'task', id: string } or null.
duration	JSON	{ start_date: date, end_date: date } or null.
reason	String	Reason provided by the applicant.
status	String	'pending', 'approved', 'rejected'.
reviewed_by	String	Admin ID who processed the application.

7. Repositorymodel

Field Name	Type	Description & Requirement Reference
_id	String	MongoDB Primary Key (Normalized).

name	<i>String</i>	Name of the document or asset.
type	<i>String</i>	'document', 'GD', 'VD'.
file_urls	<i>Array[String]</i>	Direct storage URLs (Supports multiple files).
description	<i>String</i>	Brief description of the asset.
access_control	<i>JSON</i>	Format: { is_public: boolean, allowed_roles: [], allowed_domains: [] }
uploaded_by	<i>String</i>	Admin/User ID who uploaded the asset.
uploaded_at	<i>DateTime</i>	Timestamp of upload.

8. FormModel

Field Name	Type	Description & Requirement Reference
_id	<i>String</i>	MongoDB Primary Key (Normalized).
title	<i>String</i>	Title of the dynamic form.
instructions	<i>String</i>	Form instructions.
fields	<i>Array[JSON]</i>	Definitions of input fields.
deadline	<i>DateTime</i>	Form expiration date.
created_by	<i>String</i>	Admin ID who created the form.
created_at	<i>DateTime</i>	Timestamp.

9. FormSubmissionModel

Field Name	Type	Description & Requirement Reference
_id	String	MongoDB Primary Key (Normalized).
form_id	String	Reference to FormModel.
user_id	String	Reference to Usermodel.
responses	JSON	Key-value pairs of the user's answers.
submitted_at	DateTime	Timestamp.

10. UpdatesModel

Field Name	Type	Description & Requirement Reference
_id	String	MongoDB Primary Key (Normalized).
title	String	Headline for the community update.
content	String	Full text content.
author_id	String	Admin ID.
created_at	DateTime	Timestamp.

11. ChannelModel

Field Name	Type	Description & Requirement Reference
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_id	String	MongoDB Primary Key (Normalized).
name	String	Name of the group.
type	String	'domain_group' or 'all_team_group'.
members	Array[String]	(Optional) List of user IDs.
created_at	DateTime	Timestamp of group creation.

12. ChatModel

Field Name	Type	Description & Requirement Reference
_id	String	MongoDB Primary Key (Normalized).
sender_id	String	Reference to Usermodel.
channel_id	String	Ref to ChannelModel.
text	String	Content of the chat message.
attachments	Array[JSON]	Attached files, e.g., [{ name, url }].
mentions	Array[String]	Array of user _ids tagged in the message.
timestamp	DateTime	Time the message was sent.

13. NotificationModel

Field Name	Type	Description & Requirement Reference
_id	String	MongoDB Primary Key (Normalized).

user_id	String	Reference to Usermodel (Recipient).
title	String	Brief title.
message	String	Full notification text.
type	String	Notification type categorization.
is_read	Boolean	Status of the notification.
action_link	String	URL to redirect the user upon clicking.
created_at	DateTime	Time generated.

14. AttendanceModel

Field Name	Type	Description & Requirement Reference
_id	String	MongoDB Primary Key.
meeting_id	String	Reference to MeetingModel.
user_id	String	Reference to UserModel.
is_present	Boolean	Whether the user attended.
reason_if_absent	String	Optional reason if missed.
marked_at	DateTime	Timestamp when attendance was recorded.

